数据结构基础期中题库（题型内按章节知识点分类版）

说明：题号保持原题库编号不变；只是重新调整题目顺序，便于按章节刷题。

判断题

Ch02 Algorithm Analysis(a)(b) / 复杂度与渐进分析

判2

$N(logN)^2$ is $O(N^2)$.

判5

The best "worst-case time complexity" for any algorithm that sorts by comparisons only must be $O(NlogN)$.

判8

N^2logN and $NlogN^2$ have the same speed of growth.

判13

2^N and N^N have the same speed of growth.

判15

$NlogN^2$ and $NlogN$ have the same speed of growth.

判19

N is $O(\sqrt{N}logN)$.

判24

$\log(N!) = \Omega(N \cdot \log(N))$.

Ch02 Algorithm Analysis / 排序算法性质与逆序对

判14

If there are less than 20 inversions in an integer array, then Insertion Sort will be the best method among Quick Sort, Heap Sort and Insertion Sort.

Ch03 List / 线性表、顺序存储与链式存储

判3

If the most commonly used operations are to visit a random position and to insert and delete the last element in a linear list, then sequential storage works the fastest.

判7

For a sequentially stored linear list of length N, the time complexities for query and insertion are $O(1)$ and $O(N)$, respectively.

判12

For a sequentially stored linear list of length N, the time complexities for deleting the first element and inserting the last element are $O(1)$ and $O(N)$, respectively.

判16

If a linear list is represented by a linked list, the addresses of the elements in the memory must be consecutive.

判22

If N numbers are stored in a singly linked list in increasing order, then the average time complexity for binary search is $O(\log N)$.

判29

Given two sorted lists $L1$ and $L2$, the fastest algorithm for computing $L1 \cup L2$ has time complexity $\Theta(N)$.

Ch03 Stack and Queue / 栈与队列

判4

Given the input sequence onto a stack as {1, 2, 3, ..., N}. If the first output is i, then the j-th output must be $j - i - 1$.

判9

If keys are pushed onto a stack in the order {1, 2, 3, 4, 5}, then it is impossible to obtain the output sequence {3, 4, 1, 2, 5}.

判20

In a circular queue which is implemented by an array, the `front` value must always be no larger than the `rear` value.

判30

Given that the pushing sequence of a stack is $\{1, 2, \cdots, n\}$ and popping sequence is $\{x_1, x_2, \cdots, x_n\}$. If $x_2 = n$, we can obtain 2 different possible popping sequences.

Ch04 Binary Trees / 二叉树与树的基本性质

判1

If the postorder and inorder traversal sequences of a binary tree are the same, then none of the nodes in the tree has a right child.

判21

In a tree of degree 3, we have $n_2 + n_3 \geq n_0$, where n_i is the number of degree i nodes for $0 \leq i \leq 3$.

判26

Every node in the tree is the root of some subtree.

判27

If the pre-order traversal sequence of a binary tree is `ABC`, then `CAB` cannot be its in-order traversal sequence.

Ch04 Search Tree / 二叉搜索树

判6

In a binary search tree, the keys on the same level from left to right must be in sorted (non-decreasing) order.

判23

For a binary search tree (BST), its pre-order traversal gives nodes in non-increasing order.

判28

To find 63 from a binary search tree, one possible searching sequence is $\{39, 101, 25, 80, 70, 59, 63\}$.

Ch05 Priority Queues / 堆

判25

The preorder traversal sequence of any min-heap must be in sorted (non-decreasing) order.

Ch08 Union and Find / 并查集

判18

In Union/Find algorithm, if Unions are done by size, the depth of any node must be no more than $N/2$, but not $O(logN)$.

Ch09 Graph / 图的基本性质

判10

In a directed graph, the sum of the in-degrees and out-degrees of all the vertices is twice the total number of edges.

判11

In a directed graph, the sum of the in-degrees must be equal to the sum of the out-degrees of all the vertices.

DS00 / ADT 基本概念

判17

ADT is the abbreviation for Abstract Data Type in the textbook of data structures.

单选题

Ch02 Algorithm Analysis(a)(b) / 复杂度、递推式与代码复杂度

单26

The recurrent equations for the time complexities of programs P1 and P2 are:

P1: $T(1)=1, T(N)=T(N/2)+1$;
P2: $T(1)=1, T(N)=2T(N/2)+1$;

Then the correct conclusion about their time complexities is:

A. they are both $O(logN)$
B. $O(logN)$ for P1, and $O(N)$ for P2
C. they are both $O(N)$
D. $O(logN)$ for P1, and $O(NlogN)$ for P2

单46

Let n be a non-negative integer representing the size of input. The time complexity of the following piece of code is:

```
x = 0;
while ( n >= (x+1)*(x+1) )
    x = x+1;
```

A. $O(logn)$
B. $O(n^{1/2})$
C. $O(n)$
D. $O(n^2)$

单74

Let n be a non-negative integer representing the size of input. The time complexity of the following piece of code is:

```
x = 0;
while ( n >= (x+1)*(x+1) )
    x = x+1;
```

A. $O(n)$
B. $O(n^{1/2})$
C. $O(n^2)$
D. $O(\log n)$

Ch02 Algorithm Analysis / 排序算法

单2

Which one of the following relations is correct about the extra space taken by heap sort, quick sort and merge sort?

A. heap sort $<$ merge sort $<$ quick sort
B. heap sort $>$ merge sort $>$ quick sort
C. heap sort $<$ quick sort $<$ merge sort
D. heap sort $>$ quick sort $>$ merge sort

单9

Among the following sorting methods, which one may cause that none of the elements are at their final positions before the last run begins? Assume that there are more than 2 elements to be sorted.

A. bubble sort
B. insertion sort
C. heap sort
D. quick sort

单11

Given input { 321, 156, 57, 46, 28, 7, 331, 33, 34, 63 }. Which one of the following is the result after the 1st run of the Least Signification Digit (LSD) radix sort?

A. →331→321→33→63→34→156→46→57→7→28
B. →321→331→33→63→34→156→46→57→7→28
C. →156→28→321→331→33→34→46→57→63→7
D. →57→46→28→7→33→34→63→156→321→331

单34

Among the following sorting methods, which ones will be slowed down if we store the elements in a linked structure instead of a sequential structure?

1. Insertion sort
2. Selection Sort
3. Bubble sort
4. Shell sort
5. Heap sort

A. 1 and 2 only
B. 2 and 3 only
C. 3 and 4 only
D. 4 and 5 only

单36

To sort { 8, 3, 9, 11, 2, 1, 4, 7, 5, 10, 6 } by Shell Sort, if we obtain (4, 2, 1, 8, 3, 5, 10, 6, 9, 11, 7) after the first run, and (1, 2, 3, 5, 4, 6, 7, 8, 9, 11, 10) after the second run, then the increments of these two runs must be __ , respectively.

A. 3 and 1
B. 3 and 2
C. 5 and 2
D. 5 and 3

单37

Given input { 4321, 56, 57, 46, 28, 7, 331, 33, 234, 63 }. Which one of the following is the result after the 1st run of the Least Signification Digit (LSD) radix sort?

A. →331→4321→33→63→234→56→46→57→7→28
B. →4321→331→33→63→234→56→46→57→7→28
C. →56→28→4321→331→33→234→46→57→63→7
D. →57→46→28→7→33→234→63→56→4321→331

单38

For the quicksort implementation with the left pointer stops at an element with the same key as the pivot during the partitioning, but the right pointer does not stop in a similar case, what is the running time when all keys are equal?

A. $O(logN)$
B. $O(N)$
C. $O(NlogN)$
D. $O(N^2)$

单39

For the quicksort implementation with neither the left nor the right pointer stops when an element with the same key as the pivot is found during the partitioning, what is the running time when all keys are equal?

A. $O(logN)$
B. $O(N)$
C. $O(NlogN)$
D. $O(N^2)$

单42

Given input { 4321, 56, 57, 46, 289, 17, 331, 33, 234, 63 }. Which one of the following is the result after the 1st run of the Least Signification Digit (LSD) radix sort?

A. →4321→331→33→63→234→56→46→57→17→289
B. →331→4321→33→63→234→56→46→57→17→289
C. →56→289→4321→331→33→234→46→57→63→17
D. →57→46→289→17→33→234→63→56→4321→331

Ch03 List / 线性表与链表

单10

To delete `p` from a doubly linked list, we must do:

A. `p->prior=p->prior->prior; p->prior->next=p;`
B. `p->next->prior=p; p->next=p->next->next;`
C. `p->prior->next=p->next; p->next->prior=p->prior;`
D. `p->next=p->prior->prior; p->prior=p->next->next;`

单18

To insert `s` after `p` in a doubly linked circular list, we must do:

A. `p->next=s; s->prior=p; p->next->prior=s ; s->next=p->next;`
B. `p->next->prior=s; p->next=s; s->prior=p; s->next=p->next;`
C. `s->prior=p; s->next=p->next; p->next=s; p->next->prior=s;`
D. `s->prior=p; s->next=p->next; p->next->prior=s; p->next=s;`

单44

For a sequentially stored linear list of length N, the time complexities for query and insertion are:

A. $O(1), O(1)$
B. $O(1), O(N)$
C. $O(N), O(1)$
D. $O(N), O(N)$

单55



Which one of the following is the data structure that is best represented by the above picture?

A. Circularly linked list
B. Directed graph
C. Two-dimensional array
D. Complete binary tree

单76

For two linear lists `La` and `Lb`, to link the tail of `La` with the head of `Lb`, with which one of the following data structures can we take $O(1)$ time and minimize the extra space?

A. singly linked circular list with a tail pointer
B. singly linked circular list
C. singly linked list
D. doubly linked circular list with a dummy head node

Ch03 Stack and Queue / 栈、队列、循环队列、表达式转换

单3

Given the pushing sequence of a stack as $\{1, 2, 3, 4, 5\}$. If the first number being popped out is 4, then the last one out must be:

A. 1
B. 3
C. 5
D. 1 or 5

单7

Suppose that an array of size m is used to store a circular queue. If the front position is `front` and the current size is `size`, then the rear element must be at:

A. `front+size`
B. `front+size-1`
C. `(front+size)%m`
D. `(front+size-1)%m`

单21

Suppose that an array of size m is used to store a circular queue. If the head pointer `front` and the current size variable `size` are used to represent the range of the queue instead of `front` and `rear`, then the maximum capacity of this queue can be:

A. m-1
B. m
C. m+1
D. cannot be determined

单48

Since the speed of a printer cannot match the speed of a computer, a buffer is designed to temporarily store the data from a computer so that later the printer can retrieve data in order. Then the proper structure of the buffer shall be a:

A. stack
B. queue
C. tree
D. graph

单49

Given the popping sequence of a stack as {a, b, c, d, e}. Among the following, the impossible pushing sequence is:

A. c b a e d
B. d e a c b
C. e a b c d
D. e d a b c

单57

Suppose that we are using a stack to convert the infix expression `((a + b) - c * (d / e)) + f` to the postfix expression. What is the maximum number of operators in the stack during conversion?

A. 6
B. 4
C. 5
D. 3

单60

Suppose we have a circular queue implemented on an array `a[20]`. We store the position of the first queue element in `front`, and the next position of the last queue element in `rear`. What is the size of the circular queue when `front==7` and `rear==3`?

A. 17
B. 4
C. 5
D. 16

单67

Given an empty stack `S` and an empty queue `Q`. Push elements $\{1, 2, 3, 4, 5, 6, 7\}$ one by one onto `S`. If each element that is popped from `S` is enqueued onto `Q` immediately, and if the dequeue sequence is $\{4, 5, 7, 6, 3, 2, 1\}$, then the minimum size of `S` must be:

A. 5
B. 3
C. 4
D. 2

单69

If the pushing sequence of a stack is $\{1, 2, 3, \ldots, N\}$ and the popping sequence is $\{p_1, p_2, p_3, \ldots, p_N\}$. Given that $p_1 = N$, then p_i must be:

A. i
B. $n - i$
C. $n - i + 1$
D. cannot be determined

单73

Given a stack `S` and a queue `Q`, where `S` is initialized to be empty and $\{1, 2, 3, 4, 5, 6\}$ are in `Q` (with 1 stored at the front end). If only the following three operations are allowed:

1. delete and print an element from `Q`;
2. delete an element from `Q` and push it onto `S`;
3. pop and print an element from `S`;

which of the following output sequences is NOT possible?

A. 6, 5, 4, 3, 2, 1
B. 3, 4, 5, 6, 1, 2
C. 1, 2, 5, 6, 4, 3
D. 2, 3, 4, 5, 6, 1

单75

If a stack is used to convert the infix expression `a+b*c+(d*e+f)*g` into a postfix expression, what will be in the stack (listing from the bottom up) when `f` is read?

A. `+(`
B. `abcde`
C. `++(+`
D. `+(*+`

Ch04 Binary Trees / 二叉树、完全二叉树、树的度、线索二叉树

单5

Given a tree of degree 4. Suppose that the numbers of nodes of degrees 2, 3 and 4 are 4, 2 and 1, respectively. Then the number of leaf nodes must be:

A. 8
B. 12
C. 18
D. 20

单6

Suppose that the height of a binary tree is h (the height of a leaf node is defined to be 1), and it has only the nodes of degrees 0 and 2. Then the minimum and maximum possible total numbers of nodes are:

A. $2h, 2^h - 1$
B. $2h - 1, 2^h - 1$
C. $2h - 1, 2^{h-1} - 1$
D. $2^{h-1} + 1, 2^h - 1$

单13

If on the 9th level of a complete binary tree (assume that the root is on the 1st level) there are 100 leaf nodes, then the maximum number of nodes of this tree must be:

A. 311
B. 823
C. 1847
D. cannot be determined

单17

Given a quadtree(四叉树) with 4 nodes of degree 2, 4 nodes of degree 3, 3 nodes of degree 4. The number of leaf nodes in this tree is __.

A. 22
B. 21
C. 20
D. 12

单25

In a complete binary tree with 1102 nodes, there must be __ leaf nodes.

A. 79
B. 551
C. 1063
D. cannot be determined

单29

In-order traversal of a binary tree can be done iteratively. Given the stack operation sequence as the following:

```
push(1), push(2), push(3), pop(), push(4), pop(), pop(), push(5), pop(), pop(), push(6), pop()
```

Which one of the following statements is TRUE?

A. 6 is the root
B. 2 is the parent of 4
C. 2 and 6 are siblings
D. None of the above

单30

Given a quadtree(四叉树) with 2 nodes of degree 2, 3 nodes of degree 3, 4 nodes of degree 4. The number of leaf nodes in this tree is __.

A. 10
B. 12
C. 20
D. 21

单33

How many leaf node does a complete binary tree with 2435 nodes have?

A. 1218
B. 1217
C. 812
D. cannot be determined

单35

For an in-order threaded binary tree, if the pre-order and in-order traversal sequences are

```
B E A C F D
```

and

```
A E C B D F
```

respectively, which pair of nodes' right links are both threads?

A. B and E
B. E and F
C. A and E
D. A and D

单40

For an in-order threaded binary tree, if the pre-order and in-order traversal sequences are

```
B E A C F D
```

and

`A E C B D F`

respectively, which pair of nodes' left links are both threads?

A. B and E
B. E and F
C. C and F
D. C and D

单43

Given a quadtree(四叉树) with 3 nodes of degree 2, 2 nodes of degree 3, 4 nodes of degree 4. The number of leaf nodes in this tree is __.

A. 10
B. 12
C. 20
D. 21

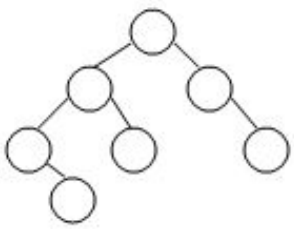
单52

A full tree of degree 3 is a tree in which every node other than the leaves has 3 children. How many leaves does a full tree of degree 3 have if it has 217 nodes?

A. 71
B. 72
C. 145
D. 146

单56

Given the shape of a binary tree shown by the figure below. If its inorder traversal sequence is {e, a, c, b, d, g, f}, then the node on the same level of `c` must be:



A. a
B. g
C. d
D. None

单58

A "full tree" of degree 3 (every non-leaf node has 3 children) has 31 nodes. Then its height h is at most ___. Note: $h = 0$ for a single node tree.

A. 10
B. cannot be determined
C. 11
D. 9

单61

In a complete binary tree with 1238 nodes, there must be ___ leaf nodes.

A. 215
B. 214
C. 620
D. 619

单64

If A and B are both leaf nodes in a binary tree, then which of the following is TRUE?

A. There exists a binary tree with pre-order traversal sequence `...A...B...` and post-order traversal sequence `...B...A...`.
B. There exists a binary tree with pre-order traversal sequence `...A...B...` and in-order traversal sequence `...B...A...`.
C. None of the above.
D. There exists a binary tree with in-order traversal sequence `...A...B...` and post-order traversal sequence `...B...A...`.

单66

A full tree of degree 4 is a tree in which every node other than the leaves has 4 children. How many leaves does a full tree of degree 4 have if it has 257 nodes?

A. 64
B. 193
C. 63
D. 194

单71

Suppose that all the leaf nodes of a given non-empty complete binary tree T are on the same level, and every non-leaf node has two children. If there are k leaf nodes in T, then T must have _ nodes in total.

A. k^2
B. $2k-1$
C. $2k$
D. 2^k-1

Ch04 Search Tree / 二叉搜索树

单1

For a binary search tree, in which order of traversal that we can obtain a non-decreasing sequence?

A. preorder traversal
B. postorder traversal
C. inorder traversal
D. level-order traversal

单23

Insert { 3, 8, 9, 1, 2, 6 } one by one into an initially empty binary search tree. The post-order traversal sequence of the resulting tree is:

A. 2, 1, 3, 6, 9, 8
B. 1, 2, 8, 6, 9, 3
C. 2, 1, 6, 9, 8, 3
D. 1, 2, 3, 6, 9, 8

单32

Insert { 5, 11, 13, 1, 3, 6 } one by one into an initially empty binary search tree. The post-order traversal sequence of the resulting tree is:

A. 3, 1, 5, 6, 13, 11
B. 3, 1, 6, 13, 11, 5
C. 1, 3, 11, 6, 13, 5
D. 1, 3, 5, 6, 13, 11

单41

Insert { 6, 9, 12, 3, 4, 8 } one by one into an initially empty binary search tree. The post-order traversal sequence of the resulting tree is:

A. 4, 3, 6, 8, 12, 9
B. 3, 4, 9, 8, 12, 6
C. 3, 4, 6, 8, 12, 9
D. 4, 3, 8, 12, 9, 6

单45

If a binary search tree of N nodes is complete, which one of the following statements is FALSE?

A. the average search time for all nodes is $O(logN)$
B. the minimum key must be at a leaf node
C. the maximum key must be at a leaf node
D. the median node must either be the root or in the left subtree

---

单63

Given a binary search tree with its level order traversal sequence $\{7, 4, 12, 3, 6, 8, 1, 5, 10\}$. If 4 is deleted from the tree, which one of the following statements is FALSE?

A. The in-order traversal is $\{1, 3, 5, 6, 7, 8, 10, 12\}$
B. 6 and 12 may be at the same level
C. 3 and 12 may be at the same level
D. 5 and 12 may be at the same level

单68

Insert $\{28, 15, 42, 18, 22, 5, 40\}$ one by one into an initially empty binary search tree. The post-order traversal sequence of the resulting tree is:

A. 5, 22, 15, 40, 18, 42, 28
B. 5, 15, 18, 22, 40, 42, 28
C. 5, 22, 18, 15, 40, 42, 28
D. 28, 22, 18, 42, 40, 15, 5

单70

Given the structure of a binary search tree (as shown in the original figure), which one of the following insertion sequences is impossible? Figure shape: root with two children; left child has two children; right child has right child.

A. 85 89 95 56 75 18
B. 85 56 89 95 18 75
C. 85 56 75 89 18 95
D. 85 89 75 56 18 95

单77

Insert $\{6, 9, 12, 3, 4, 8\}$ one by one into an initially empty binary search tree. The post-order traversal sequence of the resulting tree is:

A. 3, 4, 9, 8, 12, 6
B. 3, 4, 6, 8, 12, 9
C. 4, 3, 8, 12, 9, 6
D. 4, 3, 6, 8, 12, 9

Ch05 Priority Queues / 堆与优先队列

单4

Insert {5, 2, 7, 3, 4, 1, 6} one by one into an initially empty min-heap. The preorder traversal sequence of the resulting tree is:

A. 1, 3, 2, 5, 4, 7, 6
B. 1, 2, 3, 4, 5, 7, 6
C. 1, 2, 5, 3, 4, 7, 6
D. 1, 3, 5, 4, 2, 7, 6

单8

Suppose that the level-order traversal sequence of a min-heap is {1, 3, 2, 5, 4, 7, 6}. Use the linear algorithm to adjust this min-heap into a max-heap. The inorder traversal sequence of the resulting tree is:

A. 3, 5, 4, 2, 6, 1, 7
B. 1, 4, 3, 7, 2, 6, 5
C. 3, 5, 4, 7, 2, 6, 1
D. 4, 1, 3, 7, 6, 2, 5

单14

The result of performing three DeleteMin operations in the min-heap {1,3,2,6,7,5,4,15,14,12,9,10,11,13,8} is:

A. 4,5,6,7,8,9,10,11,12,13,14,15
B. 4,6,5,13,7,10,8,15,14,12,9,11
C. 4,6,5,12,7,10,8,15,14,9,13,11
D. 4,5,6,12,7,10,8,15,14,13,9,11

单27

The result of performing three DeleteMin operations in the min-heap {1,3,2,12,6,4,8,15,14,9,7,5,11,13,10} is:

A. 4,5,6,7,8,9,10,11,12,13,14,15
B. 4,6,5,13,7,10,8,15,14,12,9,11
C. 4,6,5,12,7,10,8,15,14,9,13,11
D. 4,5,6,12,7,10,8,15,14,13,9,11

单59

Suppose that the level-order traversal sequence of a min-heap is $\{2, 17, 5, 46, 22, 8, 10\}$. Use the linear algorithm to adjust this min-heap into a max-heap, and then call `DeleteMax`. The postorder traversal sequence of the resulting tree is: ___

A. 5, 2, 17, 8, 10, 22
B. 2, 8, 10, 5, 17, 22
C. 2, 8, 17, 5, 10, 22
D. 22, 17, 5, 2, 10, 8

单72

Using the linear algorithm to build a min-heap from the sequence $\{16, 28, 14, 13, 7, 6\}$, and then `DeleteMin`. Which one of the following statements is false?

A. 13 and 14 are siblings.
B. 7 is the root.
C. 28 is the left child of 13.
D. 13 is the parent of 16.

Ch08 Union and Find / 并查集

单12

The array representation of a disjoint set containing numbers 0 to 8 is given by { 1, -4, 1, 1, -3, 4, 4, 8, -2 }. Then to union the two sets which contain 6 and 8 (with union-by-size), the index of the resulting root and the value stored at the root are:

A. 1 and -6
B. 4 and -5
C. 8 and -5
D. 8 and -6

单22

The array representation of a disjoint set is given by { 4, 6, 5, 2, -3, -4, 3 }. If the elements are numbered from 1 to 7, the resulting array after invoking `Union(Find(7),Find(1))` with union-by-size and path-compression is:

A. { 4, 6, 5, 2, 6, -7, 3 }
B. { 4, 6, 5, 2, -7, 5, 3 }
C. { 6, 6, 5, 6, -7, 5, 5 }
D. { 6, 6, 5, 6, 6, -7, 5 }

单47

The array representation of the disjoint sets is given by { 2, –4, 2, 2, -3, 5, 6, 9, -2 }. Keep in mind that the elements are numbered from 1 to 9. After invoking `Union(Find(7), Find(9))` with union-by-size, which elements will be changed in the resulting array?

A. elements of index 5 and 7
B. elements of index 7 and 9
C. elements of index 5 and 9
D. elements of index 5, 7 and 9

单62

The array representation of the disjoint sets is given by $\{3, 1, -5, 2, 1, -3, -1, 6, 6\}$. Keep in mind that the elements are numbered from 1 to 9. After invoking `Union(Find(4), Find(8))` with union-by-size and path compression, how many elements will be changed in the resulting array?

A. 4
B. 3
C. 1
D. 2

单65

The array representation of a disjoint set containing numbers 0 to 8 is given by $\{1, -4, 1, 1, -3, 4, 4, 8, -2\}$. Then to union the two sets which contain 6 and 8 (with union-by-size), the index of the resulting root and the value stored at the root are:

A. 1 and -6
B. 8 and -6
C. 8 and -5
D. 4 and -5

Ch09 Graph Definition & Topological Sort / 图、拓扑排序、最短路、AOE

单15

If an undirected graph G = (V, E) contains 10 vertices. Then to guarantee that G is connected in any cases, there has to be at least __ edges.

A. 45
B. 37
C. 36
D. 9

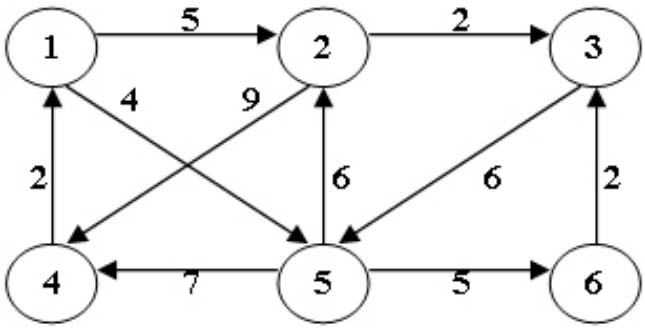
单16

What is a critical path in an AOE network?

A. the shortest circuit
B. the longest circuit
C. the shortest path from the first to the last event
D. the longest path from the first to the last event

单19

Use Dijkstra algorithm to find the shortest paths from 1 to every other vertices. In which order that the destinations must be obtained?



A. 5, 2, 3, 4, 6
B. 5, 2, 3, 6, 4
C. 5, 2, 4, 3, 6
D. 5, 2, 6, 3, 4

单20

In a weighted graph, if the length of the shortest path from `b` to `a` is 10, and there exists an edge of weight 3 between `c` and `b`, then how many of the following statements is/are TRUE?

1. The length of the shortest path from `c` to `a` must be 13.
2. The length of the shortest path from `c` to `a` must be 7.
3. The length of the shortest path from `c` to `a` must be no greater than 13.
4. The length of the shortest path from `c` to `a` must be no less than 7.

A. 1
B. 2
C. 3
D. 4

单24

Given a directed graph G=(V, E) where

$V = \{v1, v2, v3, v4, v5, v6\}$

and

$E = \{< v1, v2 >, < v1, v4 >, < v2, v6 >, < v3, v1 >, < v3, v4 >, < v4, v5 >, < v5, v2 >, < v5, v6 >\}$

Then the topological order of G is:

A. v3, v1, v4, v5, v2, v6
B. v3, v4, v1, v5, v2, v6
C. v1, v3, v4, v5, v2, v6
D. v1, v4, v3, v5, v2, v6

单28

Which of the following statements is TRUE about topological sorting?

A. If a graph has a topological sequence, then its adjacency matrix must be triangular.
B. If the adjacency matrix is triangular, then the corresponding directed graph must have a unique topological sequence.
C. In a DAG, if for any pair of distinct vertices V_i and V_j, there is a path either from V_i to V_j or from V_j to V_i, then the DAG must have a unique topological sequence.
D. If V_i precedes V_j in a topological sequence, then there must be a path from V_i to V_j.

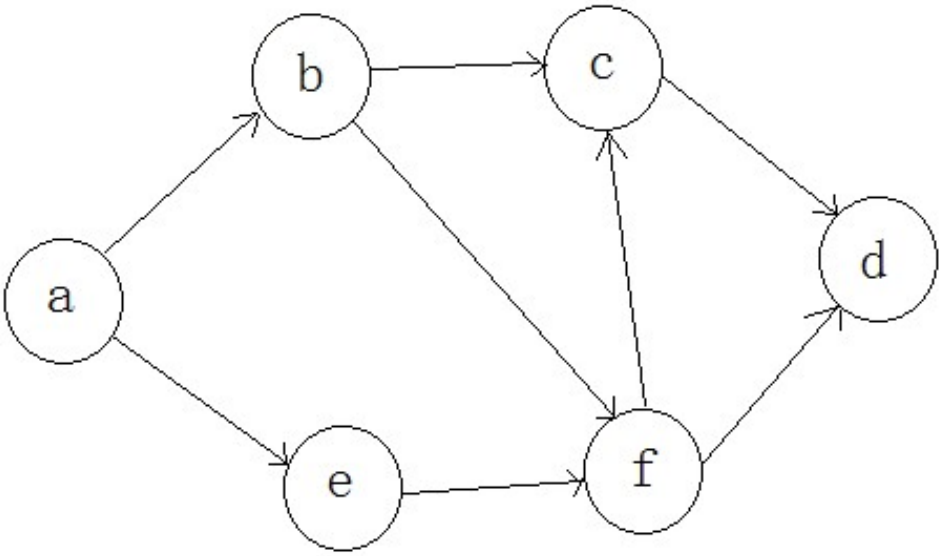
单31

If an undirected graph G = (V, E) contains 7 vertices. Then to guarantee that G is connected in any cases, there has to be at least __ edges.

A. 6
B. 15
C. 16
D. 21

单50

How many distinct topological sequences are there in the following DAG?



A. 2
B. 3
C. 4
D. 5

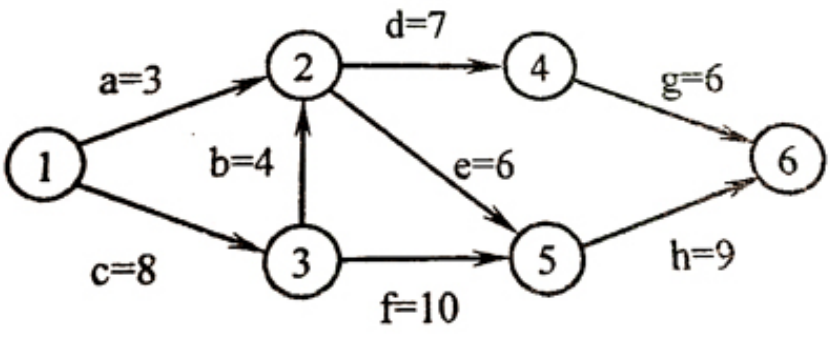
单51

If graph G is NOT connected and has 20 edges, then it must have at least __ vertices.

A. 7
B. 8
C. 9
D. 10

单53

The following figure shows the AOE network of a project with 8 activities. The earliest and the latest completion times of the activity `d` are __, respectively.



A. 3 and 7
B. 12 and 12
C. 12 and 14
D. 15 and 15

单54

If besides finding the shortest path from `S` to every other vertices, we also need to count the number of different shortest paths, we can modify the Dijkstra algorithm in the following way: add an array `count[]` so that `count[V]` records the number of different shortest paths from `S` to `V`. Then `count[V]` shall be initialized as:

A. `count[S]=1`; and `count[V]=0` for other V
B. `count[S]=0`; and `count[V]=1` for other V
C. `count[V]=1` for all vertices
D. `count[V]=0` for all vertices

程序填空题

Ch05 Priority Queues / 堆程序填空

程填1

The function is to lower the value of the integer key at position `P` by a positive amount `D` in a min-heap `H`.

```
void DecreaseKey( int P, int D, PriorityQueue H )
{
    int i, key;
    key = H->Elements[P] - D;
    for ( i = ____; H->Elements[i/2] > key; i/=2 )
        ____;
    H->Elements[i] = key;
}
```

程填3

The function is to find the `K`-th smallest element in a list `A` of `N` elements. The function `BuildMaxHeap(H, K)` is to arrange elements `H[1] ... H[K]` into a max-heap. Please complete the following program.

```
ElementType FindKthSmallest ( int A[], int N, int K )
{
    /* it is assumed that K <= N */
    ElementType *H;
```

```
    int i, next, child;

    H = (ElementType *)malloc((K+1) * sizeof(ElementType));

    for ( i = 1; i <= K; i++ )
        H[i] = A[i-1];

    BuildMaxHeap(H, K);

    for ( next = K; next < N; next++ ) {
        H[0] = A[next];

        if ( H[0] < H[1] ) {
            for ( i = 1; i * 2 <= K; i = child ) {
                child = i * 2;

                if ( child != K && ____ )
                    child++;

                if ( ____ )
                    H[i] = H[child];
                else
                    break;
            }

            H[i] = H[0];
        }
    }

    return H[1];
}
```

程填5

Please fill in the blanks in the program which deletes a given element at position `p` from a min-heap `H`.

```
Deletion ( PriorityQueue H, int p )  /* delete the element H->Elements[p] */
{
    ElementType temp;
    int child;

    temp = H->Elements[ H->Size-- ];

    if ( temp < H->Elements[p] ) {
        while ( (p != 1) && (temp < H->Elements[p/2]) ) {
            ____;
```

```
            p /= 2;
        }
    }
    else {
        while ( (child = 2*p) <= H->Size ) {
            if ( child != H->Size && ____ )
                child++;

            if ( ____ ) {
                H->Elements[p] = H->Elements[child];
                p = child;
            }
            else
                break;
        }
    }

    H->Elements[p] = temp;
}
```

Ch08 Union and Find / 并查集程序填空

程填2

Please fill in the blanks in the program which performs `Find` as a Union/Find operation with path compression.

```
SetType Find ( ElementType X, DisjSet S )
{
    ElementType root, trail, lead;

    for ( root = X; S[root] > 0; ____ ) ;

    for ( trail = X; trail != root; trail = lead ) {
        lead = S[trail];
        ____;
    }

    return root;
}
```

Ch02 Algorithm Analysis / 排序程序填空

程填4

The function is to sort the list `{ r[1] ... r[n] }` in non-decreasing order. Unlike selection sort which places only the minimum unsorted element in its correct position, this algorithm finds both the minimum and the maximum unsorted elements and places them into their final positions.

```
void sort( list r[], int n )
{
    int i, j, mini, maxi;

    for ( i = 1; i < n - i + 1; i++ ) {
        mini = maxi = i;

        for ( j = i + 1; ____; ++j ) {
            if ( ____ )
                mini = j;
            else if ( r[j]->key > r[maxi]->key )
                maxi = j;
        }

        if ( mini != i )
            swap(&r[mini], &r[i]);

        if ( maxi != n - i + 1 ) {
            if ( ____ )
                swap(&r[mini], &r[n-i+1]);
            else
                swap(&r[maxi], &r[n-i+1]);
        }
    }
}
```

Ch09 Graph / Dijkstra 程序填空

程填6

The function `Dijkstra` is to find the shortest path from `Vertex S` to every other vertices in a given `Graph`. The distances are stored in `dist[]`, and `path[]` records the paths. The `MGraph` is defined as the following:

```
typedef struct GNode *PtrToGNode;

struct GNode {
    int Nv;  /* Number of vertices */
    int Ne;  /* Number of edges    */
    WeightType G[MaxVertexNum][MaxVertexNum]; /* adjacency matrix */
};

typedef PtrToGNode MGraph;
```

Please complete the following program.

```
void Dijkstra( MGraph Graph, int dist[], int path[], Vertex S )
{
    int collected[MaxVertexNum];
    Vertex V, W;

    for ( V = 0; V < Graph->Nv; V++ ) {
        dist[V] = Graph->G[S][V];
        path[V] = -1;
        collected[V] = false;
    }

    dist[S] = 0;
    collected[S] = true;

    while (1) {
        V = FindMinDist( Graph, dist, collected );

        if ( V == ERROR )
            break;

        collected[V] = true;

        for ( W = 0; W < Graph->Nv; W++ )
            if ( collected[W] == false && Graph->G[V][W] < INFINITY ) {
                if ( ____ ) {
                    dist[W] = ____;
                    path[W] = ____;
                }
            }
    } /* end while */
}
```

Ch03 List / 链表程序填空

程填7

The function is to return the reverse linked list of `L`, with a dummy header.

```
List Reverse( List L )
{
    Position Old_head, New_head, Temp;

    New_head = NULL;
    Old_head = L->Next;
```

```
    while ( Old_head ) {
        Temp = Old_head->Next;
        ____;
        New_head = Old_head;
        Old_head = Temp;
    }

    ____;

    return L;
}
```

答案汇总（保持原题号）

判断题：

1.T 2.T 3.T 4.F 5.T
6.T 7.T 8.F 9.T 10.T
11.T 12.F 13.F 14.T 15.T
16.F 17.T 18.F 19.F 20.F
21.F 22.F 23.F 24.T 25.F
26.T 27.T 28.F 29.T 30.F

单选题：

1.C 2.C 3.D 4.D 5.B 6.B 7.D 8.C 9.B 10.C
11.B 12.B 13.B 14.B 15.B 16.D 17.A 18.D 19.B 20.B
21.B 22.D 23.C 24.A 25.B 26.B 27.C 28.C 29.C 30.D
31.C 32.B 33.A 34.D 35.D 36.B 37.B 38.D 39.D 40.D
41.D 42.A 43.C 44.B 45.C 46.B 47.C 48.B 49.B 50.A
51.B 52.C 53.C 54.A 55.A 56.B 57.C 58.A 59.A 60.D
61.D 62.A 63.B 64.C 65.D 66.B 67.A 68.C 69.C 70.D
71.B 72.C 73.B 74.B 75.A 76.A 77.C

程填1

1. P
2. H->Elements[i] = H->Elements[i/2]

程填2

1. root = S[root]
2. S[trail] = root

程填3

1. H[child+1] > H[child]
2. H[0] > H[child]

程填4

1. j <= n - i + 1
2. r[j]->key < r[mini]->key
3. maxi == i

程填5

1. H->Elements[p] = H->Elements[p/2]
2. H->Elements[child+1] < H->Elements[child]
3. temp > H->Elements[child]

程填6

1. dist[V] + Graph->G[V][W] < dist[W]
2. dist[V] + Graph->G[V][W]
3. V

程填7

1. Old_head->Next = New_head
2. L->Next = New_head